

Module 2

BIBLIOGRAPHY

ACL Reconstruction

1. Hughes et al. (2019a). Comparing the effectiveness of blood flow restriction and traditional heavy load resistance training in the post-surgery rehabilitation of anterior cruciate ligament reconstruction patients: A UK National Health Service randomised controlled trial. *Sports Medicine*. doi.org/10.1007/s40279-019-01137-2
2. Iversen et al. (2016). Intermittent blood flow restriction does not reduce atrophy following anterior cruciate ligament reconstruction. *Journal of Sport and Health Science*. 5: 115-118.
3. Kilgas et al. (2019). Exercise with blood flow restriction to improve quadriceps function long after ACL reconstruction. *Int J Sports Med*. doi: 10.1055/a-0961-1434.
4. Ohta H. (2003). Low-load resistance muscular training with moderate restriction of blood flow after anterior cruciate ligament reconstruction. *Acta Orthopaedica Scandinavica*, 74(1), 62–68. doi:10.1080/00016470310013680.
5. Takarada et al. (2000a). Effects of resistance exercise combined with moderate vascular occlusion on muscular function in humans. *J Appl Physiol*, 88: 2097-2106.
6. Takarada et al. (2000b). Applications of vascular occlusion diminish disuse atrophy of knee extensor muscles. *Med Sci Sports Exerc*. 32(12): 2035-2039.
7. Takarada et al. (2000c). Rapid increase in plasma growth hormone after low-intensity resistance exercise with vascular occlusion. *J Applied Physiol*. 88(1): 61-65.
8. Zargi et al. (2018). Short-term preconditioning with blood flow restricted exercise preserves quadriceps muscle endurance in patients after anterior cruciate ligament reconstruction. *Frontiers in Physiology*. doi: 10.3389/fphys.2018.01150.

Post Knee Arthroscopy

9. Gaunder et al. (2017). Occlusion training: pilot study for postoperative lower extremity rehabilitation following primary total knee arthroplasty. *US Army Med Dep J*.(2-17):39-43.
10. Tennent D. (2017). Blood Flow Restriction Training After Knee Arthroscopy. *Clinical Journal of Sport Medicine*. 27(3):245–252. doi:10.1097/jsm.0000000000000377

Knee Osteoarthritis

1. Ferraz et al. (2018). Benefits of resistance training with blood flow restriction in knee osteoarthritis. *Medicine & Science in Sports & Exercise*. 50(5): 897–905. doi:10.1249/mss.0000000000001530.
2. Harper et al. (2019). Blood-flow restriction resistance exercise for older adults with knee osteoarthritis: a pilot randomized clinical trial. *J Clin Med* 8(2): 265.



3. Segal NA. (2015). Efficacy of blood flow-restricted, low-load resistance training in women with risk factors for symptomatic knee osteoarthritis. *PM R*. 7:376-384. doi: <https://doi.org/10.1016/j.Pmrj.2014.09.014>.

Osteochondral Fracture

4. Loenneke JP. (2013). Rehabilitation of an osteochondral fracture using blood flow restricted exercise: a case review. *J Bodyw Mov Ther*. 2013;17:42-45. doi: <https://doi.org/10.1016/j.jbmt.2012.04.006>.

Inflammatory Myositis

5. Jørgensen (2018). Blood-flow restricted resistance training in patients with sporadic inclusion body myositis: a randomized controlled trial. *Scand J Rheumatol*, 47(5):400–409.
6. Mattar MA. (2014). Safety and possible effects of low-intensity resistance training associated with partial blood flow restriction in polymyositis and dermatomyositis. *Arthritis Res Ther*. 16:473. doi: <https://doi.org/10.1186/s13075-014-0473-5>

Muscular Weakness from Disuse

7. Cook S. (2010). Skeletal muscle adaptations following blood flow-restricted training during 30 days of muscular unloading. *J Appl Physiol* (1985).109:341-349. <https://doi.org/10.1152/japplphysiol.01288.2009>.
8. Cook et al. (2014). Improving strength and power in trained athletes with 3 weeks of occlusion training. *Int J Sports Physiol Perform*. 9: 166-172.
9. Hackney KJ. (2016). Blood flow restricted exercise compared to high load resistance exercise during unloading. *Aerosp Med Hum Perform*. 87:688-696. <https://doi.org/10.335>

Sarcopenia

10. Cook SB. (2017). Blood flow restricted resistance training in older adults at risk of mobility limitations. *Exp Gerontol*. 99:138-145. <https://doi.org/10.1016/j.exger.2017.10.004>.
11. Kim et al. (2017). Low-load resistance training with low relative pressure produces muscular changes similar to high-load resistance training. *Muscle & Nerve*, 56(6): E126–E133.
12. Vechin FC. (2015). Comparisons between low-intensity resistance training with blood flow restriction and high-intensity resistance training on quadriceps muscle mass and strength in elderly. *J Strength Cond Res*. 29:1071-1076. doi: <https://doi.org/10.1519/JSC.0000000000000703>

Achilles Tendon Rupture

13. Yow BG. (2017). Blood flow restriction training after Achilles tendon rupture. *J Foot Ankle Surg*. 57:635-638. doi: <https://doi.org/10.1053/j.Jfas.2017.11.008>.





Rheumatoid Arthritis

14. Rodrigues et al. (2019). Low-load resistance training with blood flow restriction increases muscle function, mass and functionality in women with rheumatoid arthritis. Arthritis Care & Research. Doi: 10.1002/acr.23911





Module 3 BIBLIOGRAPHY

1. Morton et al.(2018). Defining anabolic resistance. Current Opinion in Critical Care. 24(2): 124–130. doi:10.1097/mcc.0000000000000488.
2. Richter et al. (1989). Effect of exercise on insulin action in human skeletal muscle. J Appl Physiol (1985). 1989;66(2):876-885. doi:10.1152/jappl.1989.66.2.876
3. Burd et al. (2009). Exercise training and protein metabolism: influences of contraction, protein intake, and sex-based differences. J Appl Physiol. 106:1692–701.

Supplementary Resources

1. Scherbakov et al (2013). Stroke induced Sarcopenia: muscle wasting and disability after stroke. Int J Cardiol. 170(2):89-94.



Module 4

BIBLIOGRAPHY

1. Loenneke et al. (2011). Low intensity blood flow restriction training: a meta-analysis. *European Journal of Applied Physiology*, 112(5): 1849-1859.
2. Lixandrao et al. (2017). Magnitude of muscle strength and mass adaptations between high-load resistance training versus low-load resistance training associated with blood-flow restriction: A systematic review and meta-analysis. *Sports Med*, 48(2): 361-378.
3. Moritani et al. (1992). Oxygen availability and motor unit activity in humans. *Eur Journal Appl Physiol and Occup Ther*. 64(6): 552-556.
4. Manini et al. (2009). Blood flow restricted exercise and skeletal muscle health. *Exerc Sport Sci Rev*. 37(2): 78-85.
5. Takarada et al. (2000a). Effects of resistance exercise combined with moderate vascular occlusion on muscular function in humans. *J Appl Physiol*, 88: 2097-2106.
6. Takarada et al. (2000b). Applications of vascular occlusion diminish disuse atrophy of knee extensor muscles. *Med Sci Sports Exerc*. 32(12): 2035-2039.





Module 5 BIBLIOGRAPHY

1. How Does Proximity To Failure Affect Hypertrophy?

<https://medium.com/@SandCResearch/how-does-proximity-to-failure-affect-hypertrophy-e39653d41e96>

2. Does Metabolic Stress Cause Muscle Growth?

<https://medium.com/@SandCResearch/does-metabolic-stress-cause-muscle-growth-f16acd4aff41>

3. Why Is It Easy To Believe Metabolic Stress Triggers Hypertrophy (Even When It Does Not)

<https://medium.com/@SandCResearch/why-is-it-easy-to-believe-that-metabolic-stress-triggers-hypertrophy-even-when-it-does-not-2928928a431>



Module 6

BIBLIOGRAPHY

1. Gundermann et al. (2014). Activation of mTORC1 signaling and protein synthesis in human muscle following blood flow restriction exercise is inhibited by rapamycin. *Am J Physiol Endocrinol Metab* 306: E1198–E1204.
2. Fry et al. (2010). Blood flow restriction exercise stimulates mTORC1 signaling and muscle protein synthesis in older men. *J Appl Physiol*. 108:1199–1209.
3. Manini et al. (2009). Blood flow restricted exercise and skeletal muscle health. *Exerc Sport Sci Rev*. 37(2): 78-85.
4. Fujita et al. (2007). Blood flow restriction during low-intensity resistance exercise increases S6K1 phosphorylation and muscle protein synthesis. *J Appl Physiol*, 103(3): 903-910.
5. Jessee et al. (2018). Mechanisms of blood flow restriction: The new testament. *Techniques in Orthopaedics*. 33(2): 72-79. doi: 10.1097/BTO.0000000000000252
6. Lixandrao et al. (2017). Magnitude of muscle strength and mass adaptations between high-load resistance training versus low-load resistance training associated with blood-flow restriction: A systematic review and meta-analysis. *Sports Med*, 48(2): 361-378.
7. Loenneke et al. (2012). The acute muscle swelling effects of blood flow restriction. *Acta Physiologica Hungarica*, 99(4), 400–410. doi:10.1556/aphysiol.99.2012.4.4
8. Spitz et al. (2020). Blood flow restricted exercise and discomfort. *JSCR*. Volume Publish Ahead of Print. doi: 10.1519/JSC.00000000000003525
9. Amann et al. (2015). Autonomic responses to exercise: Group III/IV muscle afferents and fatigue. *Auton Neurosci* 188: 19–23.
10. Suga et al. (2010). Dose effect on intramuscular metabolic stress during low-intensity resistance exercise with blood flow restriction. *J Appl Physiol*. 108(6): 1563-1567.
11. Suga et al. (2012). Effect of multiple set on intramuscular metabolic stress during low-intensity resistance exercise with blood flow restriction. *Eur J Appl Physiol*. 112(11): 3915-3920.
12. Sugaya et al. (2011). Change in intramuscular inorganic phosphate during multiple sets of blood flow-restricted low-intensity exercise. *Clin Physiol Funct Imaging*. 31(5): 411-413.
13. Hollander et al. (2010). Partial occlusion during resistance exercise alters effort sense and pain. *J Strength Cond Res*. 24(1): 235-243.
14. Yanagisawa et al. (2018). Effects of low- load resistance exercise with blood flow restriction on intramuscular hemodynamics, oxygenation level and water content. *J Sports Med Phys Fitness* 58: 793–801.
15. Sleboda et al. (2019). Internal fluid pressure influences muscle contractile force. *Proceedings of the National Academy of Sciences*, 201914433. doi:10.1073/pnas.1914433117



16. Takarada et al. (2000). Applications of vascular occlusion diminish disuse atrophy of knee extensor muscles. *Med Sci Sports Exerc.* 32(12): 2035-2039.
17. Kubota et al (2008). Prevention of disuse muscular weakness by restriction of blood flow. *Med Sci Sports Exerc.* 40(3): 529-534.
18. Kubota et al. (2011). Blood flow restriction by low compressive force prevents disuse muscular weakness. *J Sci Med Sport.* 14(2): 95-99.
19. Barbalho et al. (2018). Addition of blood flow restriction to passive mobilization reduces the rate of muscle wasting in elderly patients in the intensive care unit: a within-patient randomized trial. *Clinical Rehabilitation.* 1-8.
20. Schliess F et al. (2006). Cell hydration and mTOR-dependent signalling. *Acta Physiol (Oxf).* 187:223-9.
21. Hamburg, et al. (2007). Physical inactivity rapidly induces insulin resistance and microvascular dysfunction in healthy volunteers. *Arteriosclerosis, thrombosis, and vascular biology*, 27(12): 2650-2656.
22. Stuart et al. (1988). Bed-rest-induced insulin resistance occurs primarily in muscle. *Metabolism.* 37:802-806.
23. Takarada et al. (2000). Rapid increase in plasma growth hormone after low-intensity resistance exercise with vascular occlusion. *J Applied Physiol.* 88(1): 61-65.
24. Manini et al. (2012). Growth hormone responses to acute resistance exercise with vascular restriction in young and old men. *Growth Horm IGF Res.* 22(5): 167-172.
25. Larkin et al. (2012). Blood flow restriction enhances post-resistance exercise angiogenic gene expression. *Med Sci Sports Exerc.* 44(11): 2077-2083.
26. Ferguson et al. (2018). The acute angiogenic signaling response to low-load resistance exercise with blood flow restriction. *Eur J Sport Sci*, 18(3): 397-406.
27. Ramis et al. (2017). Effects of acute exercise with blood flow restriction on oxidative stress markers. *Int J Sports Sci.* 7(5): 191-195.
28. Petrick et al. (2019). Blood flow restricted resistance exercise and reductions in oxygen tension attenuate mitochondrial H₂O₂ emission rates in human skeletal muscle. *The Journal of Physiology*. Doi: 10.1113/jp277765.
29. Nielsen et al. (2012). Proliferation of myogenic stem cells in human skeletal muscle in response to low-load resistance training with blood flow restriction. *J Physiol.* 590(Pt 17): 4351-4361.
30. Nielsen et al. (2017). Blood flow restricted training leads to myocellular macrophage infiltration and upregulation of heat shock proteins, but no apparent muscle damage. *J Physiol.* 595(14): 4857-4873.





31. Bjornsen et al. (2019). Delayed myonuclear addition, myofiber hypertrophy, and increases in strength with high-frequency low-load blood flow restricted training to volitional failure. J Appl Physiol. 126(3): 578-592.



Module 7

BIBLIOGRAPHY

1. Patterson et al. (2017). The role of blood flow restriction training for applied practitioners: A questionnaire-based survey. *Journal of Sports Sciences*, 36(2), 123–130. doi:10.1080/02640414.2017.1284341
2. Nakajima et al. (2006). Use and safety of KAATSU training: results of a national survey. *Int J KAATSU Training Res*. 2(1):5–13. doi: 10.3806/ijktr.2.5.
3. Poton et al. (2015). Hemodynamic responses during lower-limb resistance exercise with blood flow restriction in healthy subjects. *J Sport Med Phys Fit* 55: 1571-1577.
4. Bond et al. (2017). Restricted blood flow exercise in sedentary overweight African-american females may increase muscle strength and decrease endothelial function and vascular autoregulation. *Journal of Pharmacopuncture*, 20(1): 023-028.
5. Da Silva et al. (2019). Blood flow restriction impairs the inflammatory adaptations of strength training in overweight men: a clinical randomized trial. *Appl Physiol Nutr Metab*, doi: 10.1139/apnm-2019-0700
6. Luebbbers et al. (2017). The effects of practical blood flow restriction training on adolescent lower body strength. *Journal of Strength and Conditioning Research*. Doi: 10.1519/jsc.0000000000002302.
7. Curran et al. (2020). Blood flow restriction training applied with high-intensity exercise does not improve quadriceps muscle function after anterior cruciate ligament reconstruction: a randomized controlled trial. *Am J Sports Med*, 48(4): 825-837.
8. Chulvi-Medrano et al. (2019). Neuromuscular adaptations after blood flow restriction training combined with nutritional supplementation: a preliminary study. *Monten J Sports Sci Med* 8, 37-42.
9. Clark et al. (2011) Relative safety of 4 weeks of blood flow-restricted resistance exercise in young, healthy adults. *Scand J Med Sci Sports*. 21(5):653-662. doi:10.1111/j.1600-0838.2010.01100.x
10. Kacin et al. (2016). Safety considerations with blood flow restricted resistance training. *Annales Kinesiologiae*. 6: 3-26.
11. Mendonca et al. (2020). Nerve conduction during acute blood-flow restriction with and without low-intensity exercise nerve conduction and blood-flow restriction. *Nature*, 10: 7380.
12. Masri et al. (2020). Tourniquet-induced nerve compression injuries are caused by high pressure levels and gradients – a review of the evidence to guide surgical, pre-hospital and blood flow restriction usage. *BMC Biomed Eng*. 2(7): 1-8.



Supplementary Resources for Safety Screening:

- Bond et al. (2019). Blood Flow Restriction Resistance Exercise as a Rehabilitation Modality Following Orthopaedic Surgery: A Review of Venous Thromboembolism Risk. *J Orthop Sports Phys Ther.* 49(1):17-27.
- Brandner et al. (2018). Reported side-effects and safety considerations of blood flow restriction during exercise in practice and research. *Techniques in Orthopedics.* 33(2):1.
- da Cunha Nascimento et al. (2020). Potential Implications of Blood Flow Restriction Exercise on Vascular Health: A Brief Review. *Sports Med.* 50(1):73-81. doi:10.1007/s40279-019-01196-5
- DePhillipo et al. (2018). Blood Flow Restriction Therapy After Knee Surgery: Indications, Safety Considerations, and Postoperative Protocol. *Arthrosc Tech.* 7(10):e1037-e1043.
- DePhillipo et al. (2018). The Role of Blood Flow Restriction Therapy Following Knee Surgery: Expert Opinion. *Arthroscopy.* 34(8):2506-2510.
- Kacin et al. (2016). Safety considerations with blood flow restricted resistance training. *Annales Kinesiologiae.* 6: 3-26.
- Laswati et al. (2018). Low-Intensity Exercise with Blood Flow Restriction Increases Muscle Strength without Altering hsCRP and Fibrinogen Levels in Healthy Subjects. *Chin J Physiol.* 61(3):188-195.
- Loenneke et al. (2011). Potential safety issues with blood flow restriction training. *Scand J Med Sci Sports.* 21(4):510-518.
- Minniti et al. (2020). The Safety of Blood Flow Restriction Training as a Therapeutic Intervention for Patients With Musculoskeletal Disorders: A Systematic Review. *Am J Sports Med.* 48(7):1773-1785.
- Patterson et al. (2019). Blood Flow Restriction Exercise: Considerations of Methodology, Application, and Safety [published correction appears in *Front Physiol.* 2019 Oct 22;10:1332]. *Front Physiol.* 10:533. Published 2019 May 15. doi:10.3389/fphys.2019.00533
- Vanwyne et al. (2017). Blood Flow Restriction Training: Implementation into Clinical Practice. *International journal of exercise science.* 10(5): 649–654.
- Wernbom et al. (2020) Commentary: Can Blood Flow Restricted Exercise Cause Muscle Damage? Commentary on Blood Flow Restriction Exercise: Considerations of Methodology, Application, and Safety. *Front Physiol.* 11:243.



Module 8

BIBLIOGRAPHY

1. Loenneke et al. (2012). The acute muscle swelling effects of blood flow restriction. *Acta Physiologica Hungarica*. 99(4): 400–410. doi:10.1556/aphysiol.99.2012.4.4
2. Lisbôa et al. (2017). The time dependence of the effect of ischemic preconditioning on successive sprint swimming performance. *Journal of Science and Medicine in Sport*. 20(5): 507–511. doi:10.1016/j.jsams.2016.09.008 (IPC)
3. Arriel et al. (2019). Ischemic preconditioning delays the time of exhaustion in cycling performance during the early but not in the late phase. *Motriz: Rev Educ Fis*. 25(1): Epub ahead of print. (IPC)
4. Franz et al (2018). Ischemic preconditioning blunts muscle damage responses Induced by Eccentric Exercise. *Med Sci Sports Exerc*. 50(1):109-115. doi:10.1249/MSS.0000000000001406 (IPC)
5. Andreas et al (2011). Effect of ischemic preconditioning in skeletal muscle measured by functional magnetic resonance imaging and spectroscopy: a randomized crossover trial. *Journal of Cardiovasc Mag Res*. 13: 32. (IPC)
6. Takarada et al. (2000b). Applications of vascular occlusion diminish disuse atrophy of knee extensor muscles. *Med Sci Sports Exerc*. 32(12): 2035-2039.
7. Kubota et al (2008). Prevention of disuse muscular weakness by restriction of blood flow. *Med Sci Sports Exerc*. 40(3): 529-534.
8. Kubota et al. (2011). Blood flow restriction by low compressive force prevents disuse muscular weakness. *J Sci Med Sport*. 14(2): 95-99.
9. Barbalho et al. (2018). Addition of blood flow restriction to passive mobilization reduces the rate of muscle wasting in elderly patients in the intensive care unit: a within-patient randomized trial. *Clinical Rehabilitation*. 1-8.
10. Natsume et al. (2015) Effects of Electrostimulation with Blood Flow Restriction on Muscle Size and Strength. *Med Sci Sports Exerc*. 47(12): 2621-2627. doi:10.1249/MSS.0000000000000722
11. Santos de Oliveira Cruz et al. (2019). Effects of ischemic conditioning on maximal voluntary plantar flexion contractions. *Journal of Electromyography and Kinesiology*. doi:10.1016/j.jelekin.2019.06.004
12. Page et al. (2017). The effect of intermittent lower limb occlusion on recovery following exercise-induced muscle damage: A randomized controlled trial. *Journal of Science and Medicine in Sport*. 20(8): 729–733. doi:10.1016/j.jsams.2016.11.015 (IPOSTC)
13. Beaven et al. (2012). Intermittent lower-limb occlusion enhances recovery after strenuous exercise. *Applied Physiology, Nutrition, and Metabolism*. 37(6): 1132–1139. doi:10.1139/h2012-101 (IPOSTC)





14. Bailey et al. (2012). Effect of ischemic preconditioning on lactate accumulation and running performance. Med Sci Sports Exerc. 44(11): 2084-2089. (IPC)



Module 9

BIBLIOGRAPHY

1. Abe et al. (2006). Muscle size and strength are increased following walk training with restricted venous blood flow from the leg muscle, Kaatsu-walk training. *J Appl Physiol.* 100(5): 1460–1466. doi:10.1152/japplphysiol.01267.2005.
2. Abe et al. (2010). Effects of low-intensity cycle training with restricted leg blood flow on thigh muscle volume and VO2max in young men. *J Sports Sci Med* 9: 452.
3. Ursprung, William and Smith, John D. (2017) The Effects of Blood Flow Restriction Training on VO2Max and 1.5 Mile Run Performance. *International Journal of Exercise Science: Conference Proceedings*: 2(9): Article 108. Available at: <https://digitalcommons.wku.edu/ijesab/vol2/iss9/108>.
4. Centner et al (2019). Low-load blood flow restriction training induces similar morphological and mechanical Achilles tendon adaptations compared to high-load resistance training.
5. Amani-Shalamzari et al. (2019). Effects of blood flow restriction and exercise intensity on aerobic, anaerobic, and muscle strength adaptations in physically active collegiate women. *Front Physiol.* Doi: 10.3389/fphys.2019.00810
6. Abe et al. (2010b). Effects of low-intensity walk training with restricted leg blood flow on muscle strength and aerobic capacity in older adults. *J Geriatr Phys Ther*, 33(1): 34-40.
7. Conceicao et al. (2018). Augmented anabolic responses following 8-weeks cycling with blood flow restriction. *Med Sci Sports Ex.* doi:10.1249/MSS.0000000000000175
8. Park et al. (2010). Increase in maximal oxygen uptake following 2-week walk training with blood flow occlusion in athletes. *European Journal of Applied Physiology*.109(4): 591–600. doi:10.1007/s00421-010-1377-y.
9. Behi et al. (2016). Effect of high intensity interval training with blood flow restriction on anaerobic performance. *Int J Appl Ex Physiol.* 6(2): 45-52.
10. Törpel et al. (2018). Effect of blood flow restriction training in combination with a high-intensity interval training on physical performance. *Int J Sports Exerc Med*, 4:111.
11. Corvino et al. (2014). Four weeks of blood flow restricted training increases time to exhaustion at severe intensity cycling exercise. *J Braz Kinathrop Hum Perform* 16: 570–578.
12. Clarkson et al. (2017). Blood flow restriction walking and physical function in older adults: A randomized control trial. *J Sci Med Sport* 20: 1041–1046.
13. Clarkson et al. (2019). Chronic blood flow restricted exercise improves objective physical function: a systematic review. *Frontiers in Physiol.* Doi: 10.3389/fphys.2019.01058.
14. Spranger et al. (2015). Blood flow restriction training and the exercise pressor reflex: a call for concern. *American Journal of Physiology-Heart and Circulatory Physiology*, 309(9): H1440–H1452. doi:10.1152/ajpheart.00208.2015.



15. Christina-Oliveira et al. (2020). Clinical safety of blood flow restricted training? A comprehensive review of altered metaboreflex in cardiovascular disease during ischemic exercise. *Am J Physiol Heart Circ Physiol*, 318(1): H90-H109.
16. Estrada et al. (2020). The magnitude of the exercise pressor reflex is influenced by the active skeletal muscle mass in the decerebrate rat. *Am J Physiol Regul Integr Comp Physiol.*, 318(1): R30-37.
17. Stray-Gundersen et al. (2020). Walking with leg blood flow restriction: wide-rigid cuffs vs. narrow elastic bands. *Front Physiol*. Doi: 10.3389/fphys.2020.00568.
18. Renzi et al. (2010). Effects of leg blood flow restriction during walking on cardiovascular function. *Med. Sci. Sports Exerc.* 42, 726–732. doi: 10.1249/MSS.0b013e3181bdb454
19. Sugawara et al. (2015). Impact of leg blood flow restriction during walking on central arterial hemodynamics. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 309, R732–R739. doi: 10.1152/ajpregu.00095.2015.
20. Rossow et al. (2012). Cardiovascular and perceptual responses to blood-flow-restricted resistance exercise with differing restrictive cuffs. *Clin. Physiol. Funct. Imaging*. 32, 331–337.
21. Policarpo et al. (2019). Comparison between different prescription methods for aerobic training in young adults. *Sustainability*, 11: 6420.
22. Pescatello et al. (2014). ACSM's Guidelines for Exercise Testing and Prescription; ACSM: Indianapolis, IN, USA.
23. Swain et al. (1997). Heart rate reserve is equivalent to %VO₂reserve not to %VO₂max. *Med Sci Sports Exerc*, 29(3): 410-414.
24. Blair et al. (1989). Physical fitness and all-cause mortality. A prospective study of healthy men and women. *JAMA*, 262: 2395-2401.
25. Kodama et al. (2009). Cardiorespiratory fitness as a quantitative predictor of all-cause mortality and cardiovascular events in healthy men and women: a meta-analysis. *JAMA*, 301(19): 2024-2035.
26. Adapted from ACSM's Cardiorespiratory Fitness Guidelines – www.acsm.org
27. Formiga et al. (2020). Effect of aerobic exercise training with and without blood flow restriction on aerobic capacity in healthy young adults: a systematic review with meta-analysis. *international journal of sports physical therapy*.15(2):175–187.
28. Silva et al. (2019a). Physiological and perceptual responses to aerobic exercise with and without blood flow restriction. *J. Strength Cond. Res.* doi: 10.1519/JSC.0000000000003178. [Epub ahead of print].
29. Silva et al. (2019b). Acute and chronic responses of aerobic exercise with blood flow restriction: A systematic review. *Front Physiol*, 10:1239.
30. Mendonca et al. (2013). Metabolic cost of locomotion during treadmill walking with blood flow restriction. *Clin. Physiol. Funct. Imaging*, 34: 308–316. doi: 10.1111/cpf.12098



31. Ozaki et al. (2010). Metabolic and cardiovascular responses to upright cycle exercise with leg blood flow reduction. *J. Sports Sci. Med.* 9: 224-230.
32. Munhoz et al. (2020). Comparison of cardiopulmonary exercise testing performed with blood flow restriction vs. a traditional maximum test on execution speed, ventilatory thresholds and maximum oxygen uptake. *Sport Sciences for Health*. doi:10.1007/s11332-020-00644-4
33. Pfeiffer et al. (2019). Effects of Different Percentages of Blood Flow Restriction on Energy Expenditure. *International Journal of Sports Medicine*. 40(3): 186-190. doi:10.105
34. Karabulut et al. (2015). Hemodynamic responses and energy expenditure during blood flow restriction exercise in obese population. *Clin. Physiol. Funct. Imaging*. 37: 1–7. doi: 10.1111/cpf.12258
35. Mendonca et al. (2015). Effects of walking with blood flow restriction on excess post-exercise oxygen consumption. *Int. J. Sports Med.* 36: e11–e18. doi: 10.1055/s-0034-1395508
36. Ogawa et al. (2012). Time course changes in muscle size and fatigue during walking with restricted leg blood flow in young men. *J. Phys. Educ. Sport Manag.* 3: 14–19. doi: 10.5897/JPEM12.00.
37. Kim et al. (2015). The acute muscular effects of cycling with and without different degrees of blood flow restriction. *Acta Physiologica Hungarica*, 102(4): 428-441.
38. Ozaki et al. (2014) Effects of walking combined with restricted leg blood flow on mTOR and MAPK signaling in young men. *Acta Physiol. (Oxf)* 211: 97-106.
39. Khoubi et al. (2020) Acute interval walking with blood flow restriction could not increase ERK, p38 and decrease myostatin. *J. Sports Med. Phys. Fitness*. 60: 32-36.
40. Thomas et al. (2018). Acute physiological responses to low-intensity blood flow restriction cycling. *Journal of Sports Science and Medicine in Sport*, 21(9): 969-974.
41. Staunton et al. (2015). Haemodynamics of aerobic and resistance blood flow restriction exercise in young and older adults. *European Journal of Applied Physiology*, 115(11): 2293-2302.
42. Renzi et al. (2010). Effects of leg blood flow restriction during walking on cardiovascular function. *Med. Sci. Sports Exerc.* 42, 726–732. doi: 10.1249/MSS.0b013e3181bdb454
43. Conceicao et al. (2018). Augmented anabolic responses following 8-weeks cycling with blood flow restriction. *Med Sci Sports Ex.* doi:10.1249/MSS.000000000000175.
44. Ozaki et al. (2017) Blood flow-restricted walking in older women: does the acute hormonal response associate with muscle hypertrophy? *Clin. Physiol. Funct. Imaging* 37: 379-383.
45. Held et al. (2019). Low intensity rowing with blood flow restriction over 5-weeks increases VO₂max in elite rowers: A randomized controlled trial. *Journal of Sports Science and Medicine in Sport*.
46. Centner et al. (2018). Effects of blood flow restriction training on muscular strength and hypertrophy in older individuals: A systematic review and meta-analysis. *Sports Medicine*. doi: <https://doi.org/10.1007/s40279-018-0994-1>.





47. Sakamaki et al. (2011). Legs and trunk muscle hypertrophy following walk training with restricted leg muscle blood flow. *J. Sports Sci. Med.* 10: 338-340.
48. Patterson et al. (2019). Blood flow restriction exercise position stand: considerations of methodology, application and safety. *Frontiers in Physiol.* Doi: 10.3389.
49. De Oliveira et al. (2015). Short-term low-intensity blood flow restricted interval training improves both aerobic fitness and muscle strength. *Scand J Med Sci Sports*. DOI: 10.1111/sms/12540.



Module 10

BIBLIOGRAPHY

1. Patterson et al. (2019). Blood flow restriction exercise position stand: considerations of methodology, application and safety. *Frontiers in Physiol.* Doi: 10.3389.
2. Loenneke et al. (2014). Does blood flow restriction result in skeletal muscle damage? A critical review of available evidence. *Scand J Med Sci Sports.* 2014;24(6):e415-e422.
3. Lixandrao et al. (2017). Magnitude of muscle strength and mass adaptations between high-load resistance training versus low-load resistance training associated with blood-flow restriction: A systematic review and meta-analysis. *Sports Med,* 48(2): 361-378.
4. Nielsen et al. (2012). Proliferation of myogenic stem cells in human skeletal muscle in response to low-load resistance training with blood flow restriction. *J Physiol.* 590(Pt 17): 4351-4361.
5. Bjornsen et al. (2019). Delayed myonuclear addition, myofiber hypertrophy, and increases in strength with high-frequency low-load blood flow restricted training to volitional failure. *J Appl Physiol.* 126(3): 578-592.
6. De Lemos Muller et al (2019). Effects of low-load resistance training with blood flow restriction on the perceived exertion, muscular resistance and endurance in healthy young adults. *Sports Sciences for Health.*
7. Pignanelli et al. (2019). Low-load resistance training to task-failure with and without blood flow restriction: muscular functional and structural adaptations. *Physiology,* 318(2): R284-295.
8. Ferraz et al. (2018). Benefits of resistance training with blood flow restriction in knee osteoarthritis. *Medicine & Science in Sports & Exercise.* 50(5): 897–905.
doi:10.1249/mss.0000000000001530.
9. Rodrigues et al. (2019). Low-load resistance training with blood flow restriction increases muscle function, mass and functionality in women with rheumatoid arthritis. *Arthritis Care & Research.* Doi: 10.1002/acr.23911
10. Schoenfeld BJ. (2010). The mechanisms of muscle hypertrophy and their application to resistance training. *Journal of Str Cond Res.* 24(10): 2857-2872.
11. Franchi et al. (2017). Skeletal Muscle Remodeling in Response to Eccentric vs. Concentric Loading: Morphological, Molecular, and Metabolic Adaptations. *Front Physiol.* 8: 447.
<https://doi.org/10.3389/fphys.2017.00447>
12. Gissel H (2005). The role of Ca²⁺ in muscle cell damage. *Ann NY Acad Sci.* 1066: 166-180.
13. Fredsted et al. (2005). Anoxia induces Ca²⁺ influx and loss of cell membrane integrity in rat extensor digitorum longus muscle. *Exp Physiol.* 90(5):703-714.
14. Huerta-Alardín et al. (2005). Bench-to-bedside review: Rhabdomyolysis -- an overview for clinicians. *Critical care (London, England).* 9(2):158–169.



15. Nakajima et al. (2006). Use and safety of KAATSU training: results of a national survey. *Int J KAATSU Training Res.* 2(1):5–13. doi: 10.3806/ijktr.2.5.
16. Clark et al. (2017). Can KAATSU Exercise Cause Rhabdomyolysis? *Clinical Journal of Sport Medicine.* 27(1): e1–e2.
17. Tabata et al. (2016). Rhabdomyolysis after performing blood flow restriction training: a case report. *J Strength Cond Res* 30(7): 2064–2068.
18. Iversen et al. (2010). Low-load ischemic exercise-induced rhabdomyolysis. *Clin J Sport Med.* 20:218–219.
19. Krieger et al. (2018). A case of rhabdomyolysis caused by blood flow-restricted resistance training. *J. Spec. Oper. Med.* 18:16–17.
20. Torres et al. (2015). Rhabdomyolysis: pathogenesis, diagnosis, and treatment. *The Ochsner Journal.* 15: 58-69.
21. Cervellin et al. (2010). Rhabdomyolysis: historical background, clinical, diagnostic and therapeutic features. *Clin Chem Lab Med.* 48(6):749-756. doi:10.1515/CCLM.2010.151
22. Wernbom et al. (2019). Risk of Muscle Damage With Blood Flow–Restricted Exercise Should Not Be Overlooked, *Clin J Sports Med-* doi: 10.1097/JSM.0000000000000755.
23. Sijlacks et al. (2016). Muscle damage and repeated bout effect following blood flow restricted exercise. *Eur J Appl Physiol.* 116(3): 513–525. doi:10.1007/s00421-015-3304-8.
24. Staunton et al. (2015). Haemodynamics of aerobic and resistance blood flow restriction exercise in young and older adults. *European Journal of Applied Physiology*, 115(11): 2293-2302.
25. Madarama et al. (2013). Haemostatic and inflammatory responses to blood-flow restricted exercise in patients with ischaemic heart disease: a pilot study. *Clin Physiol Funct Imaging*, 33(1): 11-17.
26. Pinto et al. (2016). Haemodynamic responses during resistance exercise with blood flow restriction in hypertensive subjects. *Clin Physiol Funct Imaging.* doi:10.1111/cpf.12245
27. Araújo et al (2015). The effects of water-based exercise in combination with blood flow restriction on strength and functional capacity in post-menopausal women. *AGE.* 37(6). doi:10.1007/s11357-015-9851-4.
28. Pinto et al. (2018). Acute resistance exercise with blood flow restriction in elderly hypertensive women: haemodynamic, rating of perceived exertion and blood lactate. *Clin Physiol Funct I.* 38: 17-24.
29. Christina-Oliveira et al. (2020). Clinical safety of blood flow restricted training? A comprehensive review of altered metaboreflex in cardiovascular disease during ischemic exercise. *Am J Physiol Heart Circ Physiol*, 318(1): H90-H109.
30. Thomas et al. (2018a). Acute physiological responses to low-intensity blood flow restriction cycling. *Journal of Sports Science and Medicine in Sport*, 21(9): 969-974.



31. Freitas et al. (2019). Perceptual responses to continuous versus intermittent blood flow restriction exercise: a randomized controlled trial. *Physiology & Behavior*. Doi: 10.1016/j.physbeh.2019.112717
32. Freitas et al. (2020). Acute physiological responses to resistance exercise with continuous versus intermittent blood flow restriction: a randomized controlled trial. *Front Physiol*.
33. Figueroa et al. (2011). Post-exercise aortic hemodynamic responses to low-intensity resistance exercise with and without vascular occlusion. *Scand J Med Sci Sports*, 21: 431-436.
34. Fitschen et al. (2014). Perceptual effects and efficacy of intermittent or continuous blood flow restriction resistance training. *Clinical Physiology and Functional Imaging*. 34(5): 356-363.
35. Poton et al (2016). Hemodynamic response to resistance exercise with and without blood flow restriction in healthy subjects. *Clin Physiol Funct Imaging*. 36: 231–236
36. Poton et al. (2015). Hemodynamic responses during lower-limb resistance exercise with blood flow restriction in healthy subjects. *J Sport Med Phys Fit* 55: 1571-1577.
37. Vieira et al. (2013). Hemodynamic responses to resistance exercise with restricted blood flow in young and older men. *J Strength Cond Res*. 27: 2288-2294.
38. Scott et al. (2018). Hemodynamic responses to low-load blood flow restriction and unrestricted high-load resistance exercise in older women. *Front Physiol*, 9.
39. Takano et al. (2005). Hemodynamic and hormonal responses to a short-term low-intensity resistance exercise with the reduction of muscle blood flow. *Eur J Appl Physiol*. 95(1): 65-73.
40. Kacin et al. (2011). Frequent low-load ischemic resistance exercise to failure enhances muscle oxygen delivery and endurance capacity. *Scan J Med Sci Sports*. E231-241.
41. Libardi et al (2017). Hemodynamic Responses to Blood Flow Restriction and Resistance Exercise to Muscular Failure. *Int J Sports Med* 38: 134-140.
42. Narloch et al. (1995). Influence of breathing technique on arterial blood pressure during heavy weight lifting. *Arch Phys Med Rehab*. 76(5): 457-462.
43. Manini et al. (2009). Blood flow restricted exercise and skeletal muscle health. *Exerc Sport Sci Rev*. 37(2): 78-85.
44. Suga et al. (2012). Effect of multiple set on intramuscular metabolic stress during low-intensity resistance exercise with blood flow restriction. *Eur J Appl Physiol*. 112(11): 3915-3920.
45. Suga et al. (2010). Dose effect on intramuscular metabolic stress during low-intensity resistance exercise with blood flow restriction. *J Appl Physiol*. 108(6): 1563-1567.
46. Cook SB. (2017). Blood flow restricted resistance training in older adults at risk of mobility limitations. *Exp Gerontol*. 99:138-145. <https://doi.org/10.1016/j.exger.2017.10.004>.
47. Counts et al. (2016). Influence of relative blood flow restriction pressure on muscle activation and muscle adaptation. *Muscle Nerve*. 53:438–45



48. Loenneke et al. (2014). Blood flow restriction pressure recommendations: the hormesis hypothesis. *Med Hypotheses*. 82(5):623-626. doi:10.1016/j.mehy.2014.02.023
49. Fatela et al. (2016). Acute effects of exercise under different levels of blood-flow restriction on muscle activation and fatigue. *Eur J Appl Physiol*. 116(5):985-995. doi:10.1007/s00421-016-3359-1
50. Lixandrão et al. (2015). Effects of exercise intensity and occlusion pressure after 12 weeks of resistance training with blood-flow restriction. *Eur. J. Appl. Physiol*. 115:2471–80.
51. Downs et al. (2014). Acute vascular and cardiovascular responses to blood flow-restricted exercise. *Med Sci Sports Exerc*. 46(8):1489-1497.
52. Mattocks et al. (2017). The effects of upper body exercise across different levels of blood flow restriction on arterial occlusion pressure and perceptual responses. *Physiol Behav*. 171:181-186. doi:10.1016/j.physbeh.2017.01.015
53. Naess TC (2020). Determining the optimal blood flow restriction protocol for maximising muscle hypertrophy and strength, pressure and cuff width: A mini-review. *Journal of Human Sport and Exercise*, [S.l.], p. in press, may 2020.
54. Grønfeldt et al (2020). Effect of blood-flow restricted vs heavy-load strength training on muscle strength: Systematic review and meta-analysis. *Scand J Med Sci Sports*. 30(5):837-848.
55. Petrella et al. (2008). Potent myofiber hypertrophy during resistance training in humans is associated with satellite cell-mediated myonuclear addition: a cluster analysis. *J Appl Physiol* (1985). 104(6):1736-1742. doi:10.1152/japplphysiol.01215.2007
56. Bjornsen et al. (2019). Delayed myonuclear addition, myofiber hypertrophy, and increases in strength with high-frequency low-load blood flow restricted training to volitional failure. *J Appl Physiol*. 126(3): 578-592.
57. Sijl et al. (2016). Muscle damage and repeated bout effect following blood flow restricted exercise. *Eur J Appl Physiol*. 116(3): 513–525. doi:10.1007/s00421-015-3304-8.
58. Yanagisawa et al. (2017). Effects of low-load resistance exercise with blood flow restriction on high-energy phosphate metabolism and oxygenation level in skeletal muscle. *Interv. Med Appl Sci*. 9(2): 67-75. restricted exercise. *Acta Physiol (Oxf)*. 201(2): 255-263.
59. Manini et al. (2012). Growth hormone responses to acute resistance exercise with vascular restriction in young and old men. *Growth Horm IGF Res*. 22(5): 167-172.
60. Hughes et al. (2019b). Examination of the comfort and pain experienced with blood flow restriction training during post-surgery rehabilitation of anterior cruciate ligament reconstruction patients: A UK National Health Service trial. *Physical Therapy in Sport*. Doi: 10.1016/j.ptsp.2019.06.014
61. Borg et al. (2010). The Borg CR Scales® Folder. Hasselby, Sweden: Borg Perception.



62. Bell et al. (2018). Moderately heavy exercise produces lower cardiovascular, RPE, and discomfort compared to lower load exercise with and without blood flow restriction. *Eur J Appl Physiol.* 118(7):1473-1480.
63. Dankel et al. (2018). Can blood flow restriction augment muscle activation during high-load training? *Clin Physiol Funct Imaging.* 38(2): 291-295.
64. Wernbom et al. (2009). Acute effects of blood flow restriction on muscle activity and endurance during fatiguing dynamic knee extensions at low load. *J Strength Cond Res.* 23:2389-95.
65. Spitz et al. (2020). Blood Flow Restricted Exercise and Discomfort, *Journal of Strength and Conditioning Research*. Volume Publish Ahead of Print - Issue - doi: 10.1519/JSC.0000000000003525
66. Hollander et al. (2010). Partial occlusion during resistance exercise alters effort sense and pain. *J Strength Cond Res.* 24(1): 235-243.
67. Bowman et al. (2019). Proximal, distal, and contralateral effects of blood flow restriction training on the lower extremities: a randomized controlled trial. *Sports Health.* 11(2): 149-156.
68. Bowman et al. (2020). Upper-extremity blood flow restriction: the proximal, distal, and contralateral effects-a randomized controlled trial. *J Shoulder Elbow Surg.* 29(6):1267-1274. doi:10.1016/j.jse.2020.02.003
69. Yasuda et al. (2010a). Effects of low-intensity bench press training with restricted arm muscle blood flow on chest muscle hypertrophy: a pilot study. *Clinical Physiology and Functional Imaging.* doi:10.1111/j.1475-097x.2010.00949.x
70. Yamanaka et al. (2012). Occlusion training increases muscular strength in division IA football players. *J Strength Cond Res* 26: 2523–2529.
71. Luebbbers et al. (2014). The effects of a 7-week practical blood flow restriction program on well- trained collegiate athletes. *J Strength Cond Res* 28: 2270–2280.



Module 11

BIBLIOGRAPHY

1. Patterson et al. (2019). Blood flow restriction exercise position stand: considerations of methodology, application and safety. *Frontiers in Physiol.* Doi: 10.3389.
2. Hughes et al (2018). Influence and reliability of lower-limb arterial occlusion pressure at different body positions. *PeerJ.* 2018;6:e4697.
3. Sieljacks et al (2018). Body position influences arterial occlusion pressure: implications for the standardization of pressure during blood flow restricted exercise. *Eur J Appl Physiol.* 118(2):303-312.
4. Crossley et al. (2019). Effect of cuff pressure on blood flow during blood Flow–restricted rest and exercise. *Medicine & Science in Sports & Exercise*, 1. doi:10.1249/mss.0000000000002156
5. Weatherholt et al. (2019). The Effect of Cuff Width for Determining Limb Occlusion Pressure: A Comparison of Blood Flow Restriction Devices. *Int J Exerc Sci.* 12(3):136-143.
6. Mattocks et al. (2019). Perceptual changes to progressive resistance training with and without blood flow restriction. *J Sports Sci.* 37(16):1857-1864.
7. Behringer et al. (2016). Low-intensity sprint training with blood flow restriction improves 100-m dash. *JSCR.* 31(9): 2462-2472.
8. Paton et al. (2017). The effects of muscle blood flow restriction during running training on measures of aerobic capacity and run time to exhaustion. *Eur J Appl Physiol* 117: 2579–2585.
9. Amani-Shalamzari et al. (2019). Effects of Blood Flow Restriction and Exercise Intensity on Aerobic, Anaerobic, and Muscle Strength Adaptations in Physically Active Collegiate Women. *Frontiers in physiology*, 10: 810. <https://doi.org/10.3389/fphys.2019.00810>
10. Held et al. (2019). Low intensity rowing with blood flow restriction over 5-weeks increases VO2max in elite rowers: A randomized controlled trial. *Journal of Sports Science and Medicine in Sport.*
11. De Oliveira et al. (2015). Short-term low-intensity blood flow restricted interval training improves both aerobic fitness and muscle strength. *Scandinavian Journal of Medicine & Science in Sports.* 26(9):1017–1025. doi:10.1111/sms.12540.
12. Wernbom et al. (2020) Commentary: Can Blood Flow Restricted Exercise Cause Muscle Damage? Commentary on Blood Flow Restriction Exercise: Considerations of Methodology, Application, and Safety. *Front Physiol.* 11:243.
13. Spitz et al (2020). Blood Flow Restricted Exercise and Discomfort: A Review [published online ahead of print, 2020 Feb 13]. *J Strength Cond Res.* doi:10.1519/JSC.0000000000003525



14. Bjornsen et al. (2018). Type 1 muscle fiber hypertrophy after blood flow-restricted training in powerlifters. *Med & Sci in Sport Exerc.* doi: 10.1097/PSY0000000000000637
15. Sijlacks et al. (2019). Non-failure blood flow restricted exercise induces similar muscle adaptations and less discomfort than failure protocols. *Scand J Med Sci Sports.* 29: 336–347.
16. Yasuda et al (2011). Combined effects of low-intensity blood flow restriction training and high- intensity resistance training on muscle strength and size. *Eur J Appl Physiol* 111: 2525–2533.
17. Yamanaka et al (2012). Occlusion training increases muscular strength in division IA football players. *J Strength Cond Res* 26: 2523-2529.
18. Lowery et al (2014). Practical blood flow restriction training increases muscle hypertrophy during a periodized resistance training programme. *Clin Physiol Funct Imaging* 34: 317–321.
19. Hansen et al. (2020). Effects of alternating blood-flow restricted training and heavy-load resistance training on myofiber morphology and mechanical muscle function. *J Appl Physiol* (1985), doi: 10.1152/japplphysiol.00015.2020. Doi: 10.1519/JSC.00000000000003178
20. Luebbers et al. (2014). The effects of a 7-week practical blood flow restriction program on well- trained collegiate athletes. *J Strength Cond Res* 28: 2270–2280.
21. Rolnick et al. (2020). Blood flow restriction training and the physique athlete: A practical research-based guide to maximizing muscle size. *Str Cond J.* doi: 10.1519/SSC.0000000000000553
22. Minniti et al. (2019). The safety of blood flow restriction training as a therapeutic intervention for patients with musculoskeletal disorders: A systematic review. *American Journal of Sports Medicine.* doi:10.1177/0363546519882652.
23. Nascimento et al. (2019). Effects of blood flow restriction exercise on hemostasis: a systematic review of randomized and non-randomized trials. *International journal of general medicine.* 12: 91–100.
24. Cook et al. (2007). Effects of exercise load and blood-flow restriction on skeletal muscle function. *Med Sci Sports Exerc.* 39: 1708-1713.
25. Suga et al. (2010). Dose effect on intramuscular metabolic stress during low-intensity resistance exercise with blood flow restriction. *J Appl Physiol.* 108(6): 1563-1567.
26. Downs et al. (2014). Acute vascular and cardiovascular responses to blood flow-restricted exercise. *Med Sci Sports Exerc.* 46(8):1489-1497.
27. Counts et al. (2016). Influence of relative blood flow restriction pressure on muscle activation and muscle adaptation. *Muscle Nerve.* 53:438–45
28. Lixandrão et al. (2015). Effects of exercise intensity and occlusion pressure after 12 weeks of resistance training with blood-flow restriction. *Eur. J. Appl. Physiol.* 115:2471–80.



29. Mattocks et al. (2017). The effects of upper body exercise across different levels of blood flow restriction on arterial occlusion pressure and perceptual responses. *Physiol Behav.* 171:181-186. doi:10.1016/j.physbeh.2017.01.015
30. Fatela et al. (2016). Acute effects of exercise under different levels of blood-flow restriction on muscle activation and fatigue. *Eur J Appl Physiol.* 116(5):985-995. doi:10.1007/s00421-016-3359-1
31. Mouser et al. (2018). High-pressure blood flow restriction with very low load resistance training results in peripheral vascular adaptations similar to heavy resistance training. *Physiol. Meas.* doi: 10.1088/1361-6579
32. Jessee et al. (2018). Muscle adaptations to high-load training and very low-load training with and without blood flow restriction. *Frontiers in physiology.* 9: 1448.



Module 12

BIBLIOGRAPHY

1. Woo et al. (1982). Mechanical properties of tendons and ligaments. *Biorheology*, 19(3), 397–408. doi:10.3233/bir-1982-19302
2. Arya et al. (2010). Tendinopathy alters mechanical and material properties of the Achilles tendon. *J Appl Physiol*, 108: 670-675.
3. Beyer et al (2015). Heavy Slow Resistance Versus Eccentric Training as Treatment for Achilles Tendinopathy: A Randomized Controlled Trial. *Am J Sports Med*. 43(7):1704-1711. doi:10.1177/0363546515584760
4. Kongsgaard et al (2009). Corticosteroid injections, eccentric decline squat training and heavy slow resistance training in patellar tendinopathy. *Scand J Med Sci Sports*. 19(6):790-802.
5. Arampatzis A, Peper A, Bierbaum S, Albracht K. Plasticity of human Achilles tendon mechanical and morphological properties in response to cyclic strain. *J Biomech*. 2010;43(16):3073-3079. doi:10.1016/j.jbiomech.2010.08.014
6. Centner et al (2019). Low-load blood flow restriction training induces similar morphological and mechanical Achilles tendon adaptations compared to high-load resistance training.
7. Klein et al. (2002). Flexor tendon wound healing in vitro: the effect of lactate on tendon cell proliferation and collagen production *J Hand Surg Am*. 26(5):847-854.
8. Doessing et al. (2010). Growth hormone stimulates the collagen synthesis in human tendon and skeletal muscle without affecting myofibrillar protein synthesis. *The Journal of Physiology*. 588(Pt 2), 341–351.
9. Skovlund et al. (2020). The effect of low-load resistance training with blood flow restriction on chronic patellar tendinopathy – a case series. *Translational Sports Medicine*. doi:10.1002/tsm2.151
10. Sata S (2005). Kaatsu training for patella tendinitis patient. *Int J KAATSU Train Res*, 1(1): 29-32.
11. Korakakis et al. (2018). Low load resistance training with blood flow restriction decreases anterior knee pain more than resistance training alone. A pilot randomized controlled trial. DOI: 10.1016/j.ptsp.2018.09.007
12. Hughes et al. (2020). The effect of blood flow restriction exercise on exercise-induced hypoalgesia and endogenous opioid and endocannabinoid mechanisms of pain modulation. *J Appl Physiol* (1985), doi: 10.1152/jappphysiol.00768.2019
13. Hughes et al. (2019). Examination of the comfort and pain experienced with blood flow restriction training during post-surgery rehabilitation of anterior cruciate ligament reconstruction patients: A UK National Health Service trial. *Physical Therapy in Sport*. Doi: 10.1016/j.ptsp.2019.06.014



14. Hughes et al. (2019). Comparing the effectiveness of blood flow restriction and traditional heavy load resistance training in the post-surgery rehabilitation of anterior cruciate ligament reconstruction patients: A UK National Health Service randomised controlled trial. *Sports Medicine*. doi.org/10.1007/s40279-019-01137-2
15. Giles et al. (2017). Quadriceps strengthening with and without blood flow restriction in the treatment of patellofemoral pain: a double-blind randomised trial. *British Journal of Sports Medicine*, 51(23): 1688-1694.
16. Rolnick et al. (2020). Blood flow restriction training and the physique athlete: A practical research-based guide to maximizing muscle size. *JSCR*. Published Ahead of Print. doi: 10.1519/SSC.0000000000000553
17. Rittweger et al. (2006) Bone loss from the human distal tibia epiphysis during 24 days of unilateral limb suspension. *J Physiol*. 577:331–337.
18. Van der Meulen et al. (2005). Progress in understanding disuse osteopenia. *Current Opinion in Orthopaedics*. 16(5): 325 - 330. doi:10.1097/01.bco.0000178257.55565.a7
19. Hopkins et al. (2018). Disuse osteopenia following leg fracture in postmenopausal women: Implications for HIP fracture risk and fracture liaison services. *Radiography (Lond)*. 24(2):151-158.
20. Watanabe et al. (2004). Intravenous pamidronate prevents femoral bone loss and renal stone formation during 90-day bed rest. *J Bone Miner Res*. 19(11):1771-1778.
21. Narayanan et al. (2004). Differential skeletal responses of hindlimb unloaded rats on a vitamin D-deficient diet to 1,25-dihydroxyvitamin D3 and its analog, seocalcitol (EB1089). *Bone*. 35:134–143.
22. Turner CH (1998). Three rules for bone adaptation to mechanical stimuli. *Bone*. 23:399–407.
23. Tsuzuku et al. (2001). Effects of high versus low-intensity resistance training on bone mineral density in young males. *Calcif Tissue Int*. 68(6):342-347. doi:10.1007/s002230010001
24. Zhao et al. (2014). Efficiency of jumping exercise in improving bone mineral density among premenopausal women: a meta-analysis. *Sports Med*. 44(10):1393-1402. doi:10.1007/s40279-014-0220-8
25. Loenneke et al. (2012). Blood flow restriction: Rationale for improving bone. *Medical Hypotheses*. 78(4): 523–527. doi:10.1016/j.mehy.2012.01.024
26. Welch et al. (1993). Bone changes associated with intraosseous hypertension in the caprine tibia. *J Bone Joint Surg Am*. 75(1):53-60.
27. Bittar et al. (2018). Effects of blood flow restriction exercises on bone metabolism: a systematic review. *Clin Physiol Funct Imag*. 38(6): 930-935.
28. Lambert et al. (2019). Blood flow restriction therapy preserves whole limb bone and muscle following ACL reconstruction. *Orthop J Sports Med*. 7(3 Suppl2): 2325967119S00196.





29. Doma et al. (2020). Lunge exercises with blood-flow restriction induces post-activation potentiation and improves vertical jump performance. Eur J Appl Physiology. DOI: 10.1007/s00421-020-04308-6.

